# Notes: Bernard, Redding & Schott (QJE, 2011)

## Multiproduct Firms and Trade Liberalization

# Contents

# 1 Big picture

- **Question.** How do multiproduct, multidestination firms choose which products to make and export, and how do those choices respond to trade liberalization?
- **Core contribution.** The paper extends heterogeneous-firm trade theory to firms that choose both a set of products and a set of destinations. Selection occurs across firms and within firms across products.
- **Theory.** Firms draw a firm-level ability parameter and product-level attributes. Firm ability affects profitability across all products, while product attributes make some products more attractive than others inside the same firm.
- **Product scope prediction.** Lower trade costs increase export opportunities but intensify selection. Surviving firms drop their least successful products and reallocate activity toward better products.
- **Cross-country prediction.** Higher variable trade costs reduce the number of exporting firms, reduce products per exporting firm, and reduce firm-product exports, but can have ambiguous implications for average exports per firm-product.
- **Within-firm prediction.** Higher-ability firms export more products, serve more destinations, and export more of a given product to a given destination.
- **Empirical setting.** The paper uses U.S. Census transaction-level trade data and product-level U.S. manufacturing data, with special emphasis on the Canada-U.S. Free Trade Agreement (CUSFTA).
- **Main empirical result.** U.S. firms more exposed to Canadian tariff reductions reduce the number of products they produce relative to firms less exposed to the tariff cuts.
- **Additional evidence.** Gravity-style estimates show that distance and GDP affect firm extensive margins, product extensive margins, and intensive margins in ways consistent with the model.
- **Takeaway.** Trade liberalization changes the internal organization of firms. It does not only reallocate market shares across firms; it also reallocates activity within firms across products and destinations.

# 2 Incremental contribution of the paper

## 2.1 Contribution relative to Melitz-style heterogeneous-firm models

- The standard Melitz model treats firms as single-product producers, so trade liberalization reallocates market shares across firms but not within firms.
- Bernard, Redding, and Schott add an internal product-choice margin. A firm can keep operating but drop marginal products when trade costs fall.
- This changes the interpretation of productivity growth under trade liberalization: measured productivity can rise because firms change product composition, not only because low-productivity firms exit.

**Interpretation:** The paper adds an important missing margin to heterogeneous-firm trade theory: endogenous product scope inside the firm.

## 2.2 Contribution relative to empirical work on exporters

- Prior exporter studies emphasized that exporters are larger and more productive than non-exporters.
- This paper shows that exporters are also multiproduct and multidestination organizations, and that the number of products and destinations are strongly related.
- The paper uses transaction-level data to decompose exports into firm extensive margins, product extensive margins, destination extensive margins, and intensive margins.

**Interpretation:** The unit of analysis is no longer just the firm. The paper makes firm-product-country observations central to trade empirics.

### 2.3 Contribution relative to trade liberalization evidence

- The paper uses CUSFTA tariff reductions as a shock to export opportunities.
- It tests the model's prediction that exposure to tariff reductions leads firms to reduce product scope.
- This evidence links a policy shock directly to firms' internal product portfolios.

**Interpretation:** The paper contributes causal-style evidence that globalization affects within-firm organization, not only aggregate trade volumes or industry composition.

### 2.4 Contribution relative to multiproduct-firm theory

- The model allows two different product-attribute structures: common product attributes and country-specific product attributes.
- This distinction generates different predictions for hierarchies of markets and products.
- The empirical results help distinguish which margins are driven by common firm ability, product appeal, and country-specific demand.

**Interpretation:** The paper gives a flexible theoretical language for understanding why the same firm may sell different product bundles in different markets.

## 3 Model primitives and economic environment

### 3.1 Preferences

There are $J$ countries. Each country has representative consumers with CES preferences over a continuum of products:

$$U_j = \left[\int_0^1 C_{jk}^{\nu}\, dk\right]^{1/\nu}, \qquad 0 < \nu < 1. \tag{1}$$

Within each product, consumers value a continuum of differentiated varieties:

$$C_{jk} = \left[\sum_{i=1}^{J} \int_{\omega\in\Omega_{ijk}} (\lambda_{ijk}(\omega) c_{ijk}(\omega))^{\rho}\, d\omega\right]^{1/\rho}, \qquad 0 < \rho < 1. \tag{2}$$

- $i$ indexes source countries, $j$ destination countries, and $k$ products.
- $\omega$ indexes firms or varieties.
- $\lambda_{ijk}(\omega)$ is a product attribute, interpreted as consumer taste or product appeal.
- The elasticity of substitution within products is larger than the elasticity across products.

**Interpretation:** Consumers substitute more easily among firm varieties inside a product than across products. This makes product selection inside firms economically meaningful.

## 3.2 Firm entry, ability, and product attributes

Firms pay a sunk entry cost and then draw:

- a firm ability parameter $\varphi$;
- a continuum of product attributes $\lambda$.

The model has two versions:

1. **Common-product-attributes specification:** product attributes differ across products but are common across destination countries.
2. **Country-specific-product-attributes specification:** product attributes differ across both products and destination countries.

**Interpretation:** Firm ability creates selection across firms. Product attributes create selection within firms. The two specifications determine whether the same products tend to be the winners in all countries or whether product success is country-specific.

## 3.3 Production and trade costs

A firm from country $i$ producing product $k$ for market $j$ faces:

- a market fixed cost $F_{ij}$;
- a product-market fixed cost $f_{ij}$;
- iceberg trade cost $\tau_{ij} > 1$ for $i \neq j$ and $\tau_{ii} = 1$.

The firm charges a constant markup over marginal cost:

$$p_{ij}(\varphi, \lambda) = \tau_{ij} \frac{1}{\rho} \frac{w_i}{\varphi}. \tag{3}$$

**Interpretation:** Product attributes affect demand, while ability affects marginal cost. Under CES preferences, both enter revenue multiplicatively, so both can be summarized as components of profitability.

## 3.4 Product cutoffs

For each firm ability $\varphi$ and destination $j$, there is a product-attribute cutoff $\lambda^*_{ij}(\varphi)$ such that the firm supplies product $k$ to market $j$ only if

$$\lambda_{ijk} \geq \lambda^*_{ij}(\varphi). \tag{4}$$

Higher-ability firms have lower cutoffs:

$$\frac{\partial \lambda^*_{ij}(\varphi)}{\partial \varphi} < 0. \tag{5}$$

**Interpretation:** More able firms can profitably sell a wider range of products. This generates within-firm product selection.

## 3.5 Firm cutoffs and market entry

For each destination market, there is a firm ability cutoff $\varphi^*_{ij}$ such that only firms with

$$\varphi \geq \varphi^*_{ij} \tag{6}$$

serve that market. High fixed and variable trade costs make export markets require higher ability than the domestic market.

**Interpretation:** The model nests the Melitz selection margin: only high-ability firms export. It adds that exporters also choose a product range inside each destination.

# 4 Core theoretical predictions

## 4.1 Prediction 1: trade liberalization and product scope

A fall in trade costs raises export opportunities and increases competition. This causes:

- low-ability firms to exit;
- surviving firms to drop low-attribute products;
- production and exports to reallocate toward better products;
- measured firm productivity to rise through a composition effect.

**Interpretation:** The firm becomes leaner. Trade liberalization makes firms concentrate on core products.

## 4.2 Prediction 2: gravity and extensive margins

Under Pareto distributions for firm ability and product attributes, higher trade costs reduce:

- the number of exporting firms;
- the number of products exported per firm;
- firm-product exports to the destination.

The model yields gravity-like relationships in which destination GDP and distance influence trade through both firm and product extensive margins.

**Interpretation:** Aggregate gravity is not only an intensive-margin phenomenon. Much of it is produced by fewer firms and fewer firm-products serving remote or small markets.

## 4.3 Prediction 3: firm ability and multiproduct exporting

Higher firm ability implies:

- more export destinations;
- more products exported;
- higher exports of a given product to a given destination;
- a larger largest product.

**Interpretation:** Firm size, product scope, and destination scope are all manifestations of the same underlying ability draw.

## 4.4 Prediction 4: product hierarchies

In the common-product-attributes specification, firms export their best products to the most destinations. A strict hierarchy emerges: if a firm exports a smaller product to a destination, it should also export its largest product to that destination.

In the country-specific-product-attributes specification, the hierarchy is imperfect because the same product can be especially appealing in one country but not another.

**Interpretation:** The paper uses product-market hierarchies to learn whether product appeal is common across countries or destination-specific.

# 5 Empirical specifications

## 5.1 CUSFTA product-scope difference-in-differences

Firm exposure to Canadian tariff reductions is the domestic-shipment-weighted average tariff cut across the firm's 1987 industries:

$$\Delta Tariff_f = \frac{\sum_i v_{fi}^{87} \Delta Tariff_i}{\sum_i v_{fi}^{87}}, \tag{7}$$

where $v_{fi}^{87}$ is firm $f$'s 1987 domestic shipments in industry $i$.

The main difference-in-differences specification is:

$$Products_{ft} = \beta(Post_t \times Exposure_f) + \eta_f + d_t + u_{ft}. \tag{8}$$

- $Products_{ft}$ is the number of five-digit SIC products produced by firm $f$.
- $Post_t = 1$ in 1992 and 0 in 1987.
- $Exposure_f = 1$ for above-median exposure to Canadian tariff reductions.
- Firm fixed effects absorb permanent firm differences.

**Interpretation:** The coefficient $\beta$ tests whether more exposed firms reduce product scope after CUSFTA relative to less exposed firms.

## 5.2 Gravity regressions for margins of trade

The paper estimates gravity regressions for:

- export value;
- number of exporters;
- average exports per exporter;
- number of products;
- density of the firm-product-country matrix.

Typical regressors include:

$$\ln(Distance_j), \qquad \ln(GDP_j), \qquad \ln(GDPpc_j), \tag{9}$$

plus fixed effects and clustering by country.

**Interpretation:** The estimates decompose aggregate gravity into firm entry, product entry, and intensive margins.

## 5.3 Firm-level export-scope regressions

The paper relates firm export outcomes to measures of firm ability such as:

- total export value;
- exports of the firm's largest product;

- employment;
- output per worker.

Outcomes include the number of products exported, the number of countries served, and exports per firm-product-country.

**Interpretation:** The goal is to test whether high-ability firms export more products to more destinations and whether average exports per product behave as predicted under Pareto assumptions.

# 6 Identification and empirical credibility

## 6.1 CUSFTA identification

- The CUSFTA exercise exploits variation in Canadian tariff reductions across industries.
- Firm exposure is predetermined using 1987 domestic shipments.
- Firm fixed effects compare the same firms before and after the reform.
- Controls include major four-digit SIC industry and log 1987 employment.

**Interpretation:** The identifying assumption is that firms with high and low exposure to Canadian tariff cuts would have had similar product-scope changes absent CUSFTA.

## 6.2 Cross-sectional gravity identification

- The gravity tests use variation across export destinations in distance, GDP, and GDP per capita.
- The theory interprets these destination characteristics as shifting market attractiveness and trade costs.
- These regressions are not causal policy experiments; they are structural-pattern tests of the model's comparative statics.

**Interpretation:** Gravity regressions provide strong model validation because the theory predicts how different margins should respond to the same destination variables.

## 6.3 Limitations

- The model uses CES preferences and monopolistic competition, which impose tractable but restrictive revenue formulas.
- The Pareto assumptions are useful for closed forms but are approximations.
- Product attributes are not directly observed; the paper infers their role from firm-product-country patterns.
- The CUSFTA analysis focuses on manufacturing firms that survive from 1987 to 1992.
- The empirical results do not rule out managerial or organizational mechanisms outside the model.

**Interpretation:** The evidence supports the model's core margins but does not claim that every aspect of multiproduct firm behavior is explained by the parsimonious theory.

# 7 Tables and figures

## 7.1 Tables I and II: product-scope changes and gravity margins

**Read:**

- Table I shows that firms more exposed to Canadian tariff reductions reduce product scope after CUSFTA.
- The coefficient on the change in products is negative across specifications, consistent with trade liberalization making firms drop marginal products.
- Table II decomposes U.S. exports using gravity regressions.
- Distance reduces export value, the number of firms, and the number of products; GDP increases all major trade margins.
- Density, defined as the fraction of potential firm-product observations that are positive, also responds to destination characteristics.

**Economic message:** Product-scope evidence supports the within-firm selection mechanism. Gravity evidence shows that distance and market size work through extensive margins as well as intensive margins.

TABLE I

U.S. MANUFACTURING FIRM SCOPE DURING THE CANADA–U.S. FREE TRADE AGREEMENT

| | [1] | [2] | [3] |
|---|---|---|---|
| Change in products | −0.059 | −0.624 | −0.572 |
| | 0.015 | 0.101 | 0.096 |
| Change in entropy | 0.011 | 0.156 | 0.153 |
| | 0.003 | 0.026 | 0.026 |
| Firm observations | 66,472 | 66,472 | 66,472 |
| Major industry dummy variables | No | Yes | Yes |
| Log 1987 employment | No | No | Yes |

*Notes.* Table reports mean difference in noted variable between surviving firms experiencing above- and below-median changes in Canadian export opportunities between 1987 and 1992. Each cell reports the mean difference and associated standard error from a separate OLS regression. Change in products refers to change in number of five-digit SIC categories produced in the United States. Change in entropy is defined in the text. Change in export opportunities refers to the output-weighted average change in Canadian tariffs across the four-digit SIC industries produced by the firm. Robust standard errors are clustered according to firms' main four-digit SIC industry. Additional covariates are included as noted.

*Table I: CUSFTA and firm product scope*

TABLE II

GRAVITY AND THE MARGINS OF U.S. EXPORTS

| | $\ln(\text{Value}_c)$ | $\ln(\text{Avg Exports}_c)$ | $\ln(\text{Obs}_c)$ | $\ln(\text{Firms}_c)$ | $\ln(\text{Products}_c)$ | $\ln(\text{Density}_c)$ | $\ln(\text{Value}_{fpc})$ |
|---|---|---|---|---|---|---|---|
| $\ln(\text{Distance}_c)$ | −1.37 | 0.05 | −1.43 | −1.17 | −1.10 | 0.84 | −0.18 |
| | 0.17 | 0.10 | 0.17 | 0.15 | 0.15 | 0.13 | 0.080 |
| $\ln(\text{GDP}_c)$ | 1.01 | 0.23 | 0.78 | 0.71 | 0.55 | 0.48 | 0.25 |
| | 0.04 | 0.02 | 0.04 | 0.03 | 0.03 | 0.03 | 0.020 |
| Constant | 7.82 | 6.03 | 1.80 | 0.52 | 3.48 | −2.20 | 4.79 |
| | 1.83 | 1.07 | 1.81 | 1.59 | 1.55 | 1.37 | 0.64 |
| Observations | 175 | 175 | 175 | 175 | 175 | 175 | 1,878,532 |
| Fixed effects | No | No | No | No | No | No | Firm-Product |
| $R^2$ | 0.82 | 0.37 | 0.75 | 0.76 | 0.68 | 0.66 | 0.70 |

*Notes.* Table reports results of OLS regressions of U.S. export value or its components on trading-partners' GDP and great-circle distance (in kilometers) from the United States. The first six columns are country-level regressions and final column is a firm-product-country level regression. Robust standard errors are noted below each coefficient; they are adjusted for clustering by country in the final column. Data are for 2002.

*Table II: gravity and U.S. export margins*

## 7.2 Table III and Figure I: firms' extensive and intensive margins

**Read:**

- Table III shows that firms with larger export value, larger largest-product exports, and more workers export more products and to more countries.
- The number of products and number of countries are positively correlated.
- Figure I shows that the largest product rises strongly with the number of products exported, while the average product rises much less.
- The pattern is shown for all exports and for HS 84–85 products to Canada.

**Economic message:** High-ability firms are broad firms: they export more products and reach more markets. However, within-firm export distributions remain skewed, with a dominant largest product.

TABLE III

Correlation of U.S. Firms' Extensive and Intensive Margins

| | ln($Products_f$) | | | | | ln($Countries_f$) | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | [1] | [2] | [3] | [4] | [5] | [6] | [7] | [8] | [9] | [10] |
| ln(Size of Largest $Product_f$) | 0.345 | | | | | 0.329 | | | | |
| | 0.003 | | | | | 0.006 | | | | |
| ln (Size of $5^{th}$-Largest $Product_f$) | | 0.405 | | | | | | | | 0.345 |
| | | 0.004 | | | | | | | | 0.005 |
| ln($Exports_f$) | | | 0.384 | | | | | 0.347 | | |
| | | | 0.004 | | | | | 0.006 | | |
| ln($TFP_f$) | | | | 0.071 | | | | | 0.076 | |
| | | | | 0.022 | | | | | 0.022 | |
| ln($Output_f$/$Worker_f$) | | | | | 0.474 | | 0.426 | | | |
| | | | | | 0.019 | | 0.020 | | | |
| Constant | −2.300 | 0.405 | −3.022 | 1.894 | 0.436 | −2.714 | −0.797 | −3.141 | 1.292 | −1.733 |
| | 0.061 | 0.004 | 0.053 | 0.006 | 0.096 | 0.078 | 0.101 | 0.072 | 0.006 | 0.051 |
| Observations | 27,987 | 16,215 | 27,987 | 27,987 | 27,987 | 27,987 | 27,987 | 27,987 | 27,987 | 16,215 |
| $R^2$ | 0.56 | 0.50 | 0.69 | 0.13 | 0.18 | 0.55 | 0.24 | 0.60 | 0.21 | 0.53 |

*Notes.* Table reports results of firm-level OLS regressions of the log number of 10-digit HS products exported by the firm, or log the number of destination countries served by the firm, on noted covariates. All regressions include dummies for firms' main four-digit SIC industry, and robust standard errors are clustered on this dimension of the data. Results in columns 2 and 7 are restricted to firms exporting at least five products. Data are for 1997.

*Table III: correlations of extensive and intensive margins*

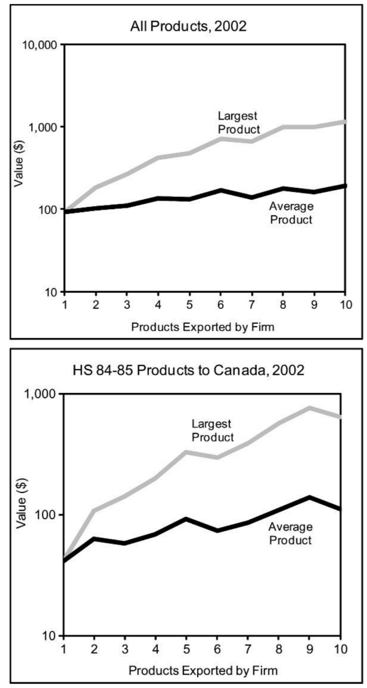


Figure I

xports of Firms' Largest and Average Products by Number of Pro

*Figure I: largest and average products by product count*

## 7.3 Table IV and Figure II: within-firm product distributions

**Read:**

- Table IV reports the share of firm exports accounted for by products at different ranks.
- The largest product accounts for about half of firm exports, while the second product accounts for about one-fifth.
- Figure II plots product rank against product size for firms exporting 10 products to Canada.
- The rank-size relationship is approximately linear on log axes, consistent with a Pareto-like distribution within firms.

**Economic message:** Firms have highly unequal product portfolios. This supports the model's product-attribute heterogeneity mechanism.

TABLE IV

DISTRIBUTION OF FIRM EXPORTS ACROSS PRODUCTS, 2002

| Rank | All Exports | Products Exported to Canada | HS 84-85 Products Exported to Canada |
|---|---|---|---|
| 1 | 49.0 | 47.4 | 47.9 |
| 2 | 18.6 | 19.4 | 19.3 |
| 3 | 10.5 | 11.1 | 11.0 |
| 4 | 6.7 | 7.0 | 7.0 |
| 5 | 4.6 | 4.8 | 4.7 |
| 6 | 3.4 | 3.4 | 3.3 |
| 7 | 2.5 | 2.5 | 2.4 |
| 8 | 1.9 | 1.9 | 1.8 |
| 9 | 1.5 | 1.5 | 1.4 |
| 10 | 1.1 | 1.1 | 1.1 |

*Table IV: distribution of firm exports across products*

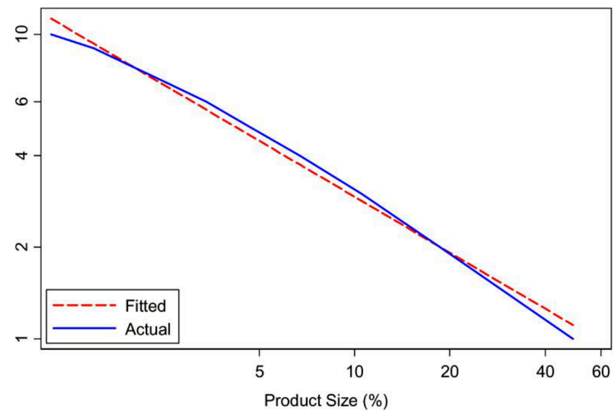


FIGURE II

Export-Product Rank versus Size (10-product exporters, 2002)

*Figure II: product rank-size relationship*

### 7.4 Table V: common and idiosyncratic components

**Read:**

- Table V decomposes firm-product-country export variation using fixed effects.
- Country-product fixed effects explain 26 percent of variation in log exports.
- Adding firm fixed effects raises explanatory power to 41 percent.
- Adding firm-product fixed effects raises the cumulative $R^2$ to 79 percent.

**Economic message:** Firm-product attributes are crucial. A large share of variation is explained by product appeal within firms, while remaining variation reflects idiosyncratic country-specific heterogeneity.

TABLE V

COMMON AND IDIOSYNCRATIC COMPONENTS OF FIRM-PRODUCT SHIPMENTS

| Fixed Effects | Cumulative $R$-Squared |
|---|---|
| Country-Product | 0.26 |
| Country-Product + Firm | 0.41 |
| Country-Product + Firm-Product | 0.79 |

*Notes.* Table reports $R^2$s of a sequence of OLS regressions of the logarithm of U.S. firm-product-country exports in 2002 on the fixed effects noted in each row.

## 8 Result synthesis

### 8.1 Trade liberalization and product pruning

The CUSFTA evidence supports the model's prediction that firms reduce product scope after trade liberalization. The key intuition is that better export opportunities increase the value of specializing in strong products and dropping weaker ones. Firms are not simply expanding in every dimension when markets open; they reorganize internally.

### 8.2 Margins of trade

The gravity evidence shows that aggregate trade declines with distance because fewer firms enter distant markets, fewer products are exported, and firm-product exports are smaller. The firm and product extensive margins are essential for understanding aggregate trade patterns.

### 8.3 Firm ability and export scope

Firms with high export value, high largest-product exports, and large employment export more products to more countries. This supports the model's view that a common firm ability component drives both product scope and market scope.

### 8.4 Product hierarchy and heterogeneity

The evidence on rank-size distributions and fixed-effect decompositions shows that product attributes are highly heterogeneous within firms. At the same time, country-specific idiosyncratic demand shocks are important enough that perfect hierarchies do not fully hold.

## 9 Short summary

- The paper develops a model of multiproduct, multidestination firms.
- Firms draw a common ability component and product-specific attributes.
- Higher-ability firms export more products, serve more destinations, and sell more of given products.
- Trade liberalization induces firms to drop low-performing products and focus on stronger products.
- CUSFTA evidence supports product pruning: firms more exposed to Canadian tariff cuts reduce product scope.
- Gravity estimates show that trade costs affect firm entry, product entry, and the intensive margin.
- Within-firm exports are highly skewed toward the largest product.
- Product-rank evidence is broadly consistent with Pareto-like product attribute distributions.
- Fixed-effect decompositions show large firm-product components and meaningful country-specific components.

## 10 Conclusion

### 10.1 Core conclusion

Bernard, Redding, and Schott show that multiproduct behavior is central to how firms respond to globalization. Trade liberalization reallocates activity both across firms and within firms, causing surviving firms to shed weaker products and focus on stronger ones.

### 10.2 Economic intuition

Firm ability creates scale and market access, while product attributes determine which products are profitable inside each firm. Opening to trade raises the return to focusing on high-attribute products and serving more attractive export markets. The result is a leaner and more productive firm product portfolio.

### 10.3 Takeaway

The paper's broad lesson is that firms are portfolios of products and markets. Trade models and trade policy analyses miss an important margin if they treat firms as single-product producers. Globalization changes the internal structure of firms, not only the distribution of activity across firms.